AI Lab-2

8-puzzle using DFS

```
PS C:\Users\Administrator\Desktop\1BM24CS316> & c:\python314\python.exe c:/Users/Administrator/Desktop/1BM24CS316/8puzzdfs.py
Initial Board Setup:
 1 2 3
 4 0 5
 7 8 6
-----------

--- Running Depth First Search (DFS) ---
DFS found a solution path in 24 moves.
 1 2 3
 4 0 5
 7 8 6
-----------
Action taken: Move blank tile 'Up'
 1 0 3
 4 2 5
 7 8 6
-----------
Action taken: Move blank tile 'Right'
 1 3 0
 4 2 5
 7 8 6
-----------
Action taken: Move blank tile 'Down'
 1 3 5
 4 2 0
 7 8 6
-----------
Action taken: Move blank tile 'Down'
 1 3 5
 4 2 6
 7 8 0
-----------
Action taken: Move blank tile 'Left'
 1 3 5
 4 2 6
 7 0 8
-----------
Action taken: Move blank tile 'Up'
 1 3 5
 4 0 6
 7 2 8
-----------
Action taken: Move blank tile 'Up'
 1 0 5
 4 3 6
 7 2 8
-----------
Action taken: Move blank tile 'Left'
 0 1 5
 4 3 6
 7 2 8
```

```
-----------
Action taken: Move blank tile 'Down'
 4 1 5
 0 3 6
 7 2 8
-----------
Action taken: Move blank tile 'Right'
 4 1 5
 3 0 6
 7 2 8
-----------
Action taken: Move blank tile 'Down'
 4 1 5
 3 2 6
 7 0 8
-----------
Action taken: Move blank tile 'Right'
 4 1 5
 3 2 6
 7 8 0
-----------
Action taken: Move blank tile 'Up'
 4 1 5
 3 2 0
 7 8 6
-----------
Action taken: Move blank tile 'Left'
 4 1 5
 3 0 2
 7 8 6
-----------
Action taken: Move blank tile 'Left'
 4 1 5
 0 3 2
 7 8 6
-----------
Action taken: Move blank tile 'Up'
 0 1 5
 4 3 2
 7 8 6
-----------
Action taken: Move blank tile 'Right'
 1 0 5
 4 3 2
 7 8 6
-----------
Action taken: Move blank tile 'Right'
 1 5 0
 4 3 2
 7 8 6
```

```
Action taken: Move blank tile 'Down'
 1 5 2
 4 3 0
 7 8 6
-----------
Action taken: Move blank tile 'Left'
 1 5 2
 4 0 3
 7 8 6
-----------
Action taken: Move blank tile 'Up'
 1 0 2
 4 5 3
 7 8 6
-----------
Action taken: Move blank tile 'Right'
 1 2 0
 4 5 3
 7 8 6
-----------
Action taken: Move blank tile 'Down'
 1 2 3
 4 5 0
 7 8 6
-----------
Action taken: Move blank tile 'Down'
 1 2 3
 4 5 6
 7 8 0
-----------
PS C:\Users\Administrator\Desktop\1BM24CS316>
```

8-puzzle using IDS

```
PS C:\Users\Administrator\Desktop\1BM24CS316> & c:\python314\python.exe c:/Users/Administrator/Desktop/1BM24CS316/8puzzids.py
Initial Board Setup:
 1 2 3
 4 0 5
 7 8 6
-----------

--- Running Iterative Deepening Search (IDS) ---
IDS found the OPTIMAL solution path in 2 moves.
 1 2 3
 4 0 5
 7 8 6
-----------
Action taken: Move blank tile 'Right'
 1 2 3
 4 5 0
 7 8 6
-----------
Action taken: Move blank tile 'Down'
 1 2 3
 4 5 6
 7 8 0
-----------
```